Customer Shopping Behavior Analysis

Analysis of 3,900 purchases across product categories to uncover spending patterns, customer segments, and strategic opportunities.

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Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Customer and purchase details

37

Missing Values

In review ratings column

Key Features

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

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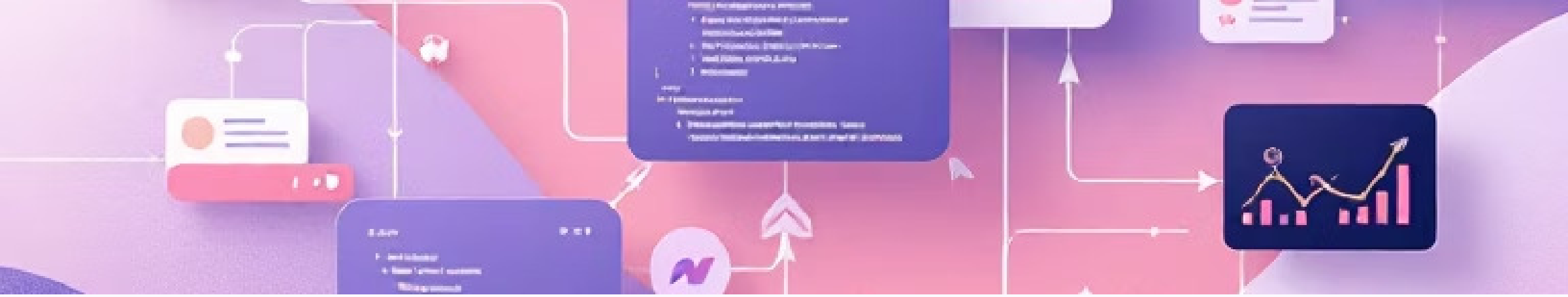

Exploratory Data Analysis: Data Preparation

Our rigorous data preparation in Python laid the foundation for accurate analysis and insights.

01

Data Loading

Imported the dataset efficiently using the `pandas` library.

02

Initial Exploration

Examined data structure with `df.info()` and summary statistics via `.describe()`.

03

Missing Data Handling

Imputed null values in 'Review Rating' using the median rating specific to each product category.

04

Column Standardization

Renamed all columns to `snake_case` for improved readability and consistency.

05

Feature Engineering

Derived 'age_group' from customer ages and 'purchase_frequency_days' from purchase dates.

06

Data Consistency

Removed redundant 'promo_code_used' column, as 'discount_applied' captured the necessary information.

07

Database Integration

Successfully loaded the prepped DataFrame into PostgreSQL for subsequent SQL analysis.

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Revenue by Gender

$157,890

Male Customers

Generated 68% of total revenue

$75,191

Female Customers

Generated 32% of total revenue

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Top 5 Products by Rating

3.86

Gloves

3.86 average rating

3.84

Sandals

3.84 average rating

3.82

Boots

3.82 average rating

3.80

Hat

3.80 average rating

3.78

Skirt

3.78 average rating

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Shipping & Subscription Analysis

Shipping Comparison

Standard: $58.46 average purchase

Express: $60.48 average purchase

Express shipping customers spend ~3.5% more per transaction.

Subscription Status

Subscribers (1,053): $59.49 avg spend, $62,645 total revenue

Non-Subscribers (2,847): $59.87 avg spend, $170,436 total revenue

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Discount Strategy Insights

Highest Discount Rate

Hat: 50.00% of purchases discounted

High Discount Products

Sneakers: 49.66% | **Coat:** 49.07% | **Sweater:** 48.17% | **Pants:** 47.37%

High-Spending Discount Users

839 customers used discounts but spent above average purchase amount ($59.76)

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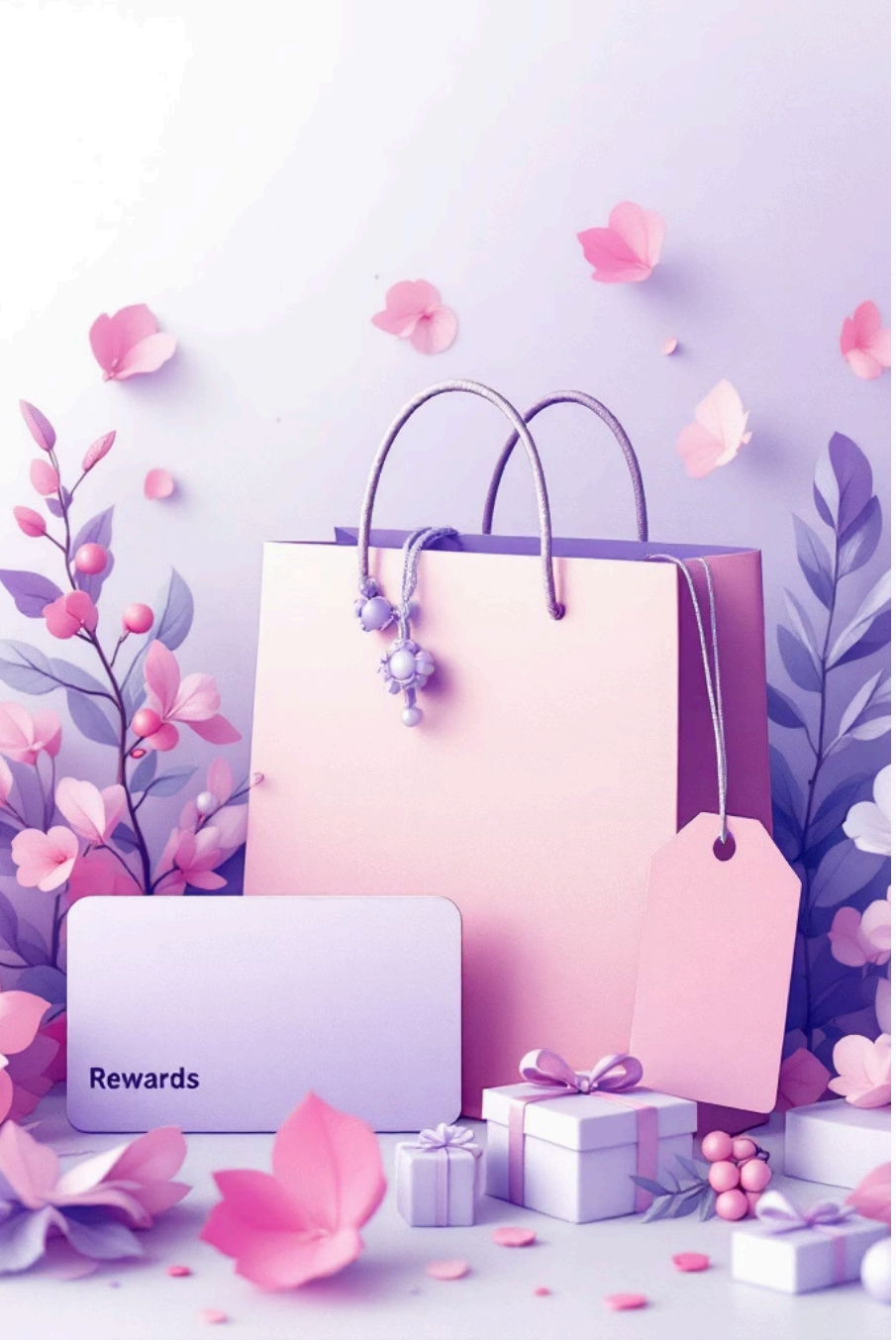

Customer Segmentation

Loyal

3,116 customers with consistent purchase history

Returning

701 customers with multiple purchases

New

83 customers with single purchase

Repeat Buyer Analysis

958 repeat buyers (over 5 purchases) vs. 2,518 single-time buyers

Subscription Conversion

Repeat buyers show higher subscription rates, indicating loyalty program potential

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Top Products by Category

Accessories

Jewelry: 171 orders | **Sunglasses:** 161 orders | **Belt:** 161 orders

Clothing

Blouse: 171 orders | **Pants:** 171 orders | **Shirt:** 169 orders

Footwear

Sandals: 160 orders | **Shoes:** 150 orders | **Sneakers:** 145 orders

Outerwear

Jacket: 163 orders | **Coat:** 161 orders

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Revenue by Age Group

Young Adult

$62,143 total revenue

Middle-aged

$59,197 total revenue

Adult

$55,978 total revenue

Senior

$55,763 total revenue

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers to increase conversion rates

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control across product categories

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users

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